

Name: Your Name

Date: May 11, 2025

Course:

1. Introduction

This report demonstrates the configuration of a secure network environment using Cisco Packet Tracer. Key security configurations include SSH, Port Security, and Access Control Lists (ACLs).

2. Network Configuration Overview

- Secure SSH access configured on R1
- Port Security on S2 (Fa0/1) to limit access to two MAC addresses
- ACL 10 (Standard) to allow only Laptop0 to access R1 via SSH
- EXAMPLE_ACL (Extended) on R2 to manage traffic to the Web, File, and DNS Servers

3. IP Address Assignment

Devices were assigned IP addresses and subnet masks as follows:

- Laptop0: 192.168.10.10 /24, GW: 192.168.10.1
- Laptop1: 192.168.10.11 /24, GW: 192.168.10.1
- R1 Gi0/1: 192.168.10.1 /24
- R1 Gi0/0/0: 97.15.5.1 /30
- R2 Gi0/0/0: 97.15.5.2 /30
- R2 Gi0/1: 87.74.49.1 /24
- Web Server: 87.74.49.7 /24, GW: 87.74.49.1
- File Server: 87.74.49.8 /24, GW: 87.74.49.1
- DNS Server: 87.74.49.9 /24, GW: 87.74.49.1

4. Key Configurations

****Router R1 Configuration:****

```
hostname R1
ip domain-name EXAMPLE
username admin password cisco
line vty 0 15
transport input ssh
login local
enable secret cisco
crypto key generate rsa
access-list 10 permit 192.168.10.10
line vty 0 15
access-class 10 in
```

****Router R2 Configuration:****

```
hostname R2
enable secret cisco
ip access-list extended EXAMPLE_ACL
permit tcp 192.168.10.0 0.0.0.255 host 87.74.49.7 eq 80
deny ip 192.168.10.0 0.0.0.255 host 87.74.49.7
deny tcp 192.168.10.0 0.0.0.255 host 87.74.49.8 eq 80
permit ip 192.168.10.0 0.0.0.255 host 87.74.49.8
deny ip 192.168.10.0 0.0.0.255 host 87.74.49.9
permit udp 192.168.10.0 0.0.0.255 host 87.74.49.9 eq 53
interface Gi0/0/0
ip access-group EXAMPLE_ACL in
```

****S2 Switch Configuration (Port Security):****

```
interface fa0/1
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address 0002.4A68.CD5C
```

switchport port-security mac-address 0007.ec6c.0e01

switchport port-security violation restrict

5. Test Results

- Ping from Laptop1 to Laptop0: Successful
- Ping from Laptop0 to WEB server: Blocked (by ACL)
- Ping from Laptop0 to File server: Successful
- Ping from Laptop1 to File server: Blocked (by ACL)
- Telnet from Laptop0 to R1: Blocked (VTY only allows SSH)
- SSH from Laptop0 to R1: Successful
- Ping from Laptop1 to R1: Successful

6. Conclusion

This configuration ensures secure remote management, limits access through port security, and implements granular access control using both standard and extended ACLs. The network was verified through pings and remote access testing.

```
R2
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
exit
R2(config)#interface gig0/1
R2(config-if)#ip address 87.74.49.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
R2(config)#ip access-list extended EXAMPLE_ACL
R2(config-ext-nacl)#Rule 1: Allow HTTP from 192.168.10.0/24 to Web Browser
R2(config-ext-nacl)#permit tcp 192.168.10.0 0.0.0.255 host 87.74.49.7 eq 80
R2(config-ext-nacl)#! Rule 2: Deny all other IP to Web Browser
R2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 host 87.74.49.7
R2(config-ext-nacl)#! Rule 3: Deny all other IP to Web Server
R2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 host 87.74.49.7
R2(config-ext-nacl)#Rule 3: Allow all except HTTP to File Server
R2(config-ext-nacl)#Rule 3: Allow all except HTTP to File Server
R2(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 87.74.49.8 eq 80
R2(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 87.74.49.8 eq 80
R2(config-ext-nacl)#Rule 3: Allow all except HTTP to File Server
R2(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 87.74.49.8 eq 80
R2(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 host 87.74.49.8
R2(config-ext-nacl)#Rule 4: Allow only DNS to DNS Server
R2(config-ext-nacl)#permit udp 192.168.10.0 0.0.0.255 host 87.74.49.9 eq 53
R2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 host 87.74
R2(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.255 host 87.74.49.9
R2(config-ext-nacl)#! Allow everything else (optional cleanup rule)
R2(config-ext-nacl)#permit ip any any
R2(config-ext-nacl)#exit
R2(config)#interface gig0/0/0
R2(config-if)#ip access-group EXAMPLE_ACL in
R2(config-if)#exit
R2(config)#
```

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable secret cisco
R1(config)#interface gig0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
R1(config)#interface gig0/0/0
R1(config-if)#ip address 97.15.5.1 255.255.255.252
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to down
R1(config-if)#exit
R1(config)#
R1(config)#ip domain-name EXAMPLE
R1(config)#username admin secret cisco
R1(config)#crypto key generate rsa
The name for the keys will be: R1.EXAMPLE
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, Keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 11:53:31.195: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 15
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
R1(config)#access-list 10 permit 192.168.10.10
R1(config)#line vty 0 15
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

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Laptop1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 87.74.49.9

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::230:A3FF:FE72:142B

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

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Laptop0

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.10.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

87.74.49.9

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::202:4AFF:FE68:CD5C

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

WEB Server

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

87.74.49.7

Subnet Mask

255.255.255.0

Default Gateway

87.74.49.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

FE80::201:63FF:FE50:72CB

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

File Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

87.74.49.8

Subnet Mask

255.255.255.0

Default Gateway

87.74.49.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::206:2AFF:FEBE:99B6

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

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DNS server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

87.74.49.9

Subnet Mask

255.255.255.0

Default Gateway

87.74.49.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::240:BFF:FE5C:93DB

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

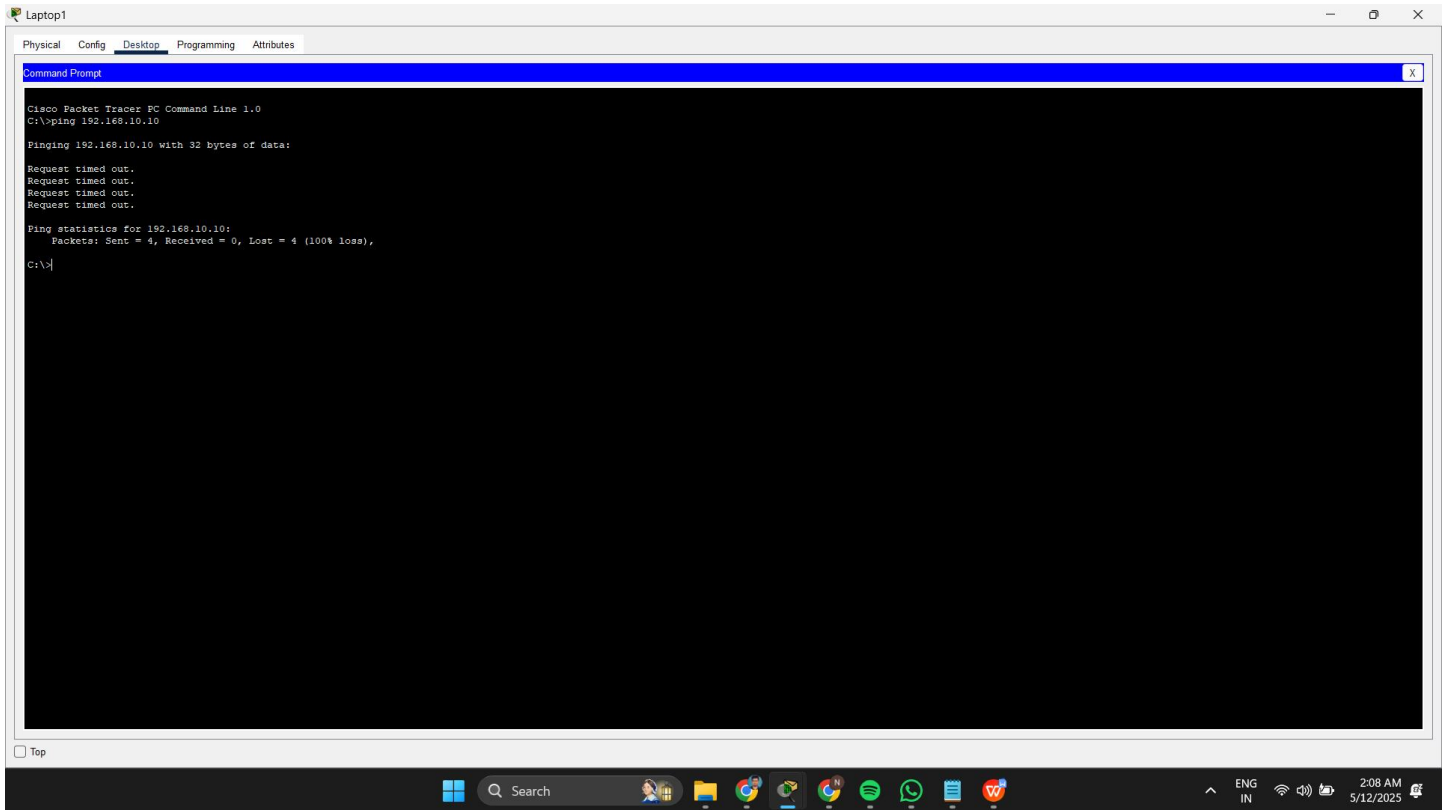
Password

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File Edit Options View Tools Extensions Window Help

Activity Results

Time Elapsed: 02:40:17

Congratulations Guest! You completed the activity.

Overall Feedback [Assessment Items](#) Connectivity Tests

[Expand/Collapse All](#) [Show Incorrect Items](#)

Assessment Items	Status	Points	Component(s)	Feedback
Network				
R1				
Ports				
GigabitEthernet0/0	Correct	0	Other	
Port Status	Correct	5	Inter-VLAN Routin...	
GigabitEthernet0/10	Correct	0	Other	
802.1Q	Correct	0	Other	
VLAN ID	Correct	5	Inter-VLAN Routin...	
IP Address	Correct	5	Inter-VLAN Routin...	
Subnet Mask	Correct	5	Inter-VLAN Routin...	
GigabitEthernet0/30	Correct	0	Other	
802.1Q	Correct	0	Other	
VLAN ID	Correct	5	Inter-VLAN Routin...	
IP Address	Correct	5	Inter-VLAN Routin...	
Subnet Mask	Correct	5	Inter-VLAN Routin...	
S1				
Ports				
FastEthernet0/11	Correct	0	Other	
Access VLAN	Correct	5	VLAN Configuration	
FastEthernet0/6	Correct	0	Other	
Access VLAN	Correct	5	VLAN Configuration	
GigabitEthernet0/1	Correct	0	Other	
Port Mode	Correct	5	Trunking Configur...	
VLANs				
VLAN 10	Correct	0	Creating a VLAN	
VLAN Name	Correct	5	VLAN Configuration	
VLAN 30	Correct	0	Creating a VLAN	
VLAN Name	Correct	5	VLAN Configuration	

Score	Item Count
60/60	12/12

Component	Items/Total	Score
Inter-VLAN Routing Configuration	7/7	35/35
Trunking Configuration	1/1	5/5
VLAN Configuration	4/4	20/20

Close

